A Thermal Modeling Method for Power Capacitors Under Long Timescale Dynamic Operation Profiles

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***Abstract*—This article proposes a thermal modeling method for film capacitors, which is applicable for high-power converters with long timescale dynamic operation profiles, such as traction and wind power systems. The proposed frequency domain modeling analyzes the effects of power loss and ambient temperature comprehensively in realistic mission profiles. Furthermore, the transfer function is discretized to concurrently obtain the iterative model with temperature rise by the influence of power loss and ambient temperature. This method has two advantages, as follows: First, discrete realistic operating profiles, including power loss and ambient temperature at different sampling times, can be conveniently introduced into the thermal model simultaneously. Second, the effect of temperature-dependent equivalent series resistance and thermal resistance can be further considered in discrete frequency domain modeling. The full power experiments for capacitor stress testing demonstrate the reasonableness and accuracy of the proposed method in hotspot temperature and lifetime expectancy calculations.**

***Index Terms*—Ambient temperature, film capacitors, high-power converter, hotspot temperature, mission profile, power loss.**

I. Introduction

CAPACITOR is one of the highest failure rate components in power electronic systems [1]. The degradation and failure of capacitors cause serious catastrophes to the converter systems [2]. The hotspot temperature of the capacitors is a critical indicator for the operation of its converter system [3]. On the one hand, the hotspot temperature of the capacitor is a key parameter in the evaluation of the capacitor lifetime bottleneck, and high hotspot temperature can significantly shorten the lifetime of the capacitors [4]. In addition, for film capacitors, the excessive hotspot temperature directly melts the film layer, leading to rapid failure of the capacitor [5].

In recent years, several solutions for thermal modeling of capacitors have been suggested, which include mainly time-domain-based thermal modeling methods. The equivalent series resistance (ESR) and power loss are used to calculate the temperature rise, which is combined with the ambient temperature to directly estimate the hotspot temperature of the capacitor [6], [7]. In [8], a simplified thermal modeling method for capacitor analysis is proposed to optimize the computational burden of thermal coupling. An ESR-temperature rise solution process is proposed in [9], where the estimated hotspot temperature is solved by iterative correction of temperature rise. However, the methods in [6], [7], [8], and [9] are only applicable to steady-state thermal calculations on short timescales as the dynamics of the operating conditions of converter level are not considered. Due to the linear relationship between capacitance and hotspot temperature, an indirect calculation method by updating the capacitance-temperature coefficient is proposed [10]. However, this method requires accurate real-time acquisition of the capacitance tolerance and still has limitations in high-power converter applications. Some commercial capacitor manufacturers provide empirical models for hotspot temperature, but do not consider the influence of dynamic power losses and ambient temperature in realistic operating systems [11], [12]. To evaluate the thermal stress of capacitors in the realistic application system, the mission profiles should not be ignored, and its temperature is affected by a combination of internal operating conditions and external ambient temperature. Considering the mission profiles, the authors in [13] and [14] proposed thermal modeling methods for dynamic power losses from the perspective of photovoltaic and railway traction systems, respectively. Based on the thermal modeling of dynamic power losses, the effect of internal resistance variation with temperature on power loss is considered [15], [16]. However, these time-domain based methods in [13], [14], [15], and [16] only consider the transfer effects of power loss on thermal modeling alone, but ignore the effect of ambient temperature on the thermal network.

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TABLE I
COMPARISON OF THERMAL MODELING METHODS FOR CAPACITORS IN CONVERTER SYSTEMS

	Calculation principle	Advantage	Limitation Is it suitable for long timescale converter operation?		Exp. error	Ref.
Time-domain model	Power loss model	Simple calculation process	Limited to constant operating conditions; Ignore the effects of ambient temperature dynamics and internal resistance	×	N/A	[6] [7]
	Thermal coupling model	Reduce the computing burden of thermal coupling		×	4%	[8]
	Simplified power loss model	Iterative correction with temperature rise		×	N/A	[9]
	Temperature coefficient model	Calculated indirectly using the temperature coefficient and case temperature		×	8%	[10]
	Empirical model of manufacturers	Substitute parameters for direct calculation	Limited to specific types and constant operating conditions	×	N/A	[11][12]
	Thermal model of dynamic power losses	Consider variations in operating conditions	Ignore the effects of ambient temperature dynamics	✓	10%	[13][14]
		Consider the effect of internal resistance		✓	8%	[15][16]
	Goertzel training model	Algorithm-based prediction by capacitance value	Ignore the dynamic effects of ambient temperature and internal resistance; Require extensive testing to build the initial database	×	6%	[21]
	Gradient-based training model	Lower computation time and training burden		×	11%	[22]
	LSTM training model	Consider load variations in the model		✓	5%	[23]
	RLS training model	Mutual update of power loss and temperature	Additional Rogowski coil	✓	N/A	[24]
Finite element model	Finite element model based on physical structure	Consider the winding structure of capacitors	cm cm Limited to constant operating conditions and require extensive computational time	×	10%	[25]
		Consider the multiple subunits of capacitors		×	N/A	[26]
		Consider the multiscale finite element model		×	3%	[27]
	Data-driven ANN model	High precision and suitable for various operating conditions	Require extensive testing to build the initial database	×	4%	[28]
Frequency-domain model	Continuous-time frequency domain model	Consider the effects of operating conditions and ambient temperature individually	From the perspective of power semiconductors	✓	N/A	[29]
			Effect of variations in converter operating conditions not considered	✓	11%	[30]
	Discrete-time frequency domain-model	Considering the combined effects of converter operating conditions, ambient temperature, and internal resistance variations	Suitable for high-power converter long timescale operation	✓	3%	Proposed

ANN：Artificial Neural Network.

In power electronic systems, parameter evaluation and monitoring methods based on digital twins and AI algorithms are gradually being applied [17], [18]. For capacitors, a few data twin-based circuit analysis modeling approaches have evaluated capacitance values and ESR [19], [20]. Focusing on hotspot temperature, Sundararajan et al. [21] proposed a Goertzel-based training model that predicts capacitance values and solves for hotspot temperature through algorithmic prediction. Rambetius [22] utilized iterative training toolbox in MATLAB to reduce computational time and training burden for the algorithm, but those methods are constrained by constant operating conditions. A temperature evaluation model considering load variations is proposed in [23] through long short-term memory (LSTM) training. Based on the recursive least squares (RLS) training model, mutual updating of power loss and hotspot temperature under dynamic operating conditions is implemented, but its needs additional rogowski coil circuits [24]. In general, those method based on digital twins and AI algorithms require extensive testing to build the initial database, and ignore the dynamic effects of ambient temperature and internal resistance.

Another thermal modeling approach is established through finite element method (FEM) simulation. The FEM thermal model considering the capacitor geometry and multisubunit structure are, respectively, analyzed in [25] and [26]. Based on FEM simulation, a multispatial-scale modeling approach is considered, achieving higher temperature estimation accuracy [27]. However, those methods still requires a significant amount of computational time for the simulation even for a single operating condition. After introducing a dataset of FEM, an evaluation algorithm of the capacitor temperature based on a neural network algorithm is proposed, which takes into account changes in operating conditions [28]. Nevertheless, this method requires a large amount of initial data for training according to finite element simulations, and the prediction range of dynamic conditions is affected by the bandwidth of the dataset.

A frequency-domain-based thermal modeling idea is proposed for power semiconductor devices to consider the effect of power loss and ambient temperature on the thermal network [29], which provides a new inspiration for the thermal modeling of capacitors. In [30], a frequency-domain thermal modeling method that considers the dynamic ambient temperature is further proposed, which is expected to be further applied to dynamic operating profiles. This method focuses on the thermal impedance frequency-domain response model, and is limited to the study from the perspective of ambient temperature effects. However, the dynamic operation profiles in converter systems need to consider the effects of power loss and ambient temperature on the response model simultaneously. Furthermore, it remains a challenge to consider the dynamic effects of temperature-dependent internal resistance and thermal resistance in the frequency-domain transfer model.

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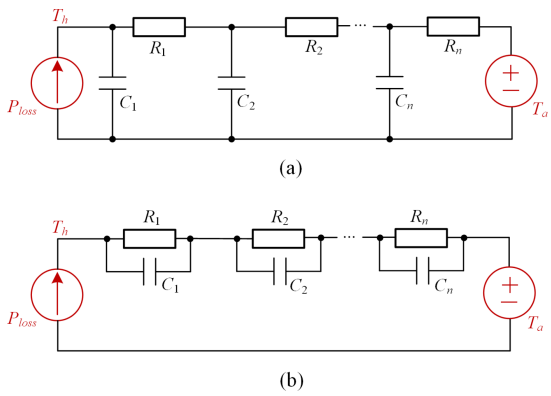


Fig. 1. Equivalent thermal modeling of capacitors with Cauer model and Foster model. (a) Equivalent thermal modeling with Cauer model. (b) Equivalent thermal modeling with Foster model.

Table I summarizes the working principles, advantages, and limitations of the existing thermal modeling methods for capacitors. Based on the above challenges, this article proposes a thermal modeling method for film capacitors, which is applicable for high-power converter systems with long timescale dynamic operation profiles. It has the following contributions.

1) The proposed frequency-domain modeling analyzes the effects of power loss and ambient temperature comprehensively in realistic mission profiles to obtain the iterative model with temperature rise.
2) By discretization of the frequency-domain model, the dynamic operating curves at any sampling time can be conveniently substituted in the thermal model, and the effect of the temperature-dependent internal resistance and thermal resistance is considered.
3) A high-power capacitor dynamic stress emulation platform is used to experimentally verify the accuracy of the proposed method compared to two representative existing methods.

The rest of this article is organized as follows. Section II presents the thermal modeling based on the frequency domain. Section III gives the proposed discretized long timescale thermal modeling. Section IV discusses the experimental verification. Finally, Section V concludes this article.

II. Thermal Modeling for Capacitors

A. Existing Thermal Modeling Wtih Time Domain

The heat of the capacitor is mainly generated by the power loss, which is closely related to the ESR and current of the capacitor. The hotspot temperature of the capacitor T_h in one conventional thermal model can be characterized as [6], [7]:

$$T_h = T_a + I_{\text{rms}}^2 \times r_s \times R_{h-a} \tag{1}$$

where T_a and R_{h-a} represent the ambient temperature of the capacitor and thermal resistance from hotspot to ambient, respectively. I_{rms} and r_s are the rms value of the capacitor current and the ESR, respectively.

Fig. 1 illustrates the equivalent thermal modeling of capacitors. Considering the dynamic effect of temperature change, the hotspot temperature of the capacitor is affected by the thermal resistance in addition to the thermal capacitance, and its thermal modeling can be expressed as [8], [9]

$$T_h = T_a + I_{\text{rms}}^2 \times r_s \times \sum_{m=1}^{k} R_m \left(1 - e^{\frac{t}{R_m \times C_m}}\right) \tag{2}$$

where R_m and C_m represent the thermal resistance and thermal capacitance of layer m, respectively. t and k are the time of system operation and the total number of layers in the thermal model, respectively.

Considering the realistic operating conditions, both the ambient temperature $T_a(t)$ and the current $I_{\text{rms}(t)}$ of the capacitor are dynamic parameters that vary with time t. The thermal modeling considering the mission profile is given in [13], [14], and [15]

$$T_h(t) = T_a(t) + \sum_{t=1}^{n} \left[I_{\text{rms}}^2(t) \times r_s \times \sum_{m=1}^{k} R_m \left(e^{-\frac{(n-t)T}{R_m \times C_m}} - e^{-\frac{(n+1-t)T}{R_m \times C_m}} \right) \right] \tag{3}$$

where n and T represent the number of discrete points and discrete intervals in the mission profiles, respectively.

The time-domain thermal model can effectively analyze the influence of thermal network parameters on power loss. Through iterative computation, the coupled effect of power loss and hotspot temperature under dynamic operating conditions can be further derived. However, the time-domain thermal model does not account for the impact of thermal network parameters on the propagation from ambient temperature to hotspot temperature. When the ambient temperature is constant, the influence of thermal network parameters can be neglected. Under long-timescale operation with dynamically varying ambient temperature, the existing time-domain thermal model still has limitations. In such cases, the effect of thermal network parameters on the ambient-to-hotspot temperature transfer is overlooked, leading to inaccuracies in the calculated hotspot temperature.

B. Thermal Modeling With Frequency Domain

In order to simultaneously consider the effects of different operation conditions (including dynamic power losses and dynamic ambient temperature) on the hotspot temperature of the capacitor, the thermal network is expanded through the frequency-domain model.

In this part, the frequency-domain thermal models under the Cauer modeling and Foster modeling are analyzed as examples of two-layer thermal networks, respectively.

1) Thermal Modeling Based on Cauer Model: Fig. 2(a) illustrates the second-order Cauer thermal modeling with the power loss of the capacitor as the input source. The transfer function from the power loss source P_{loss} to the hotspot temperature T_h in the Cauer thermal network is denoted as $G_{P_{\text{loss}}}^{\text{Cauer}}(s)$. It can be derived by successively combining the transfer functions of each *RC* branch.

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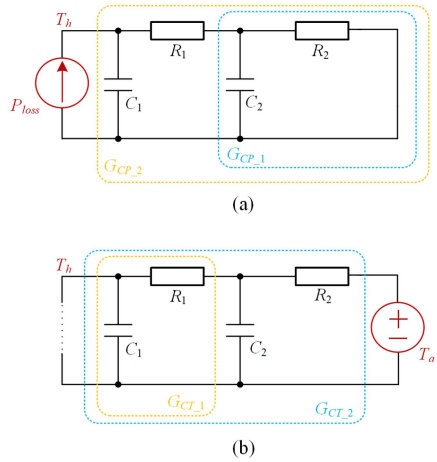


Fig. 2. Equivalent second-order thermal modeling with Cauer model with power loss source and ambient temperature source. (a) 2nd-order Cauer thermal modeling with power loss source. (b) 2nd-order Cauer thermal modeling with ambient temperature source.

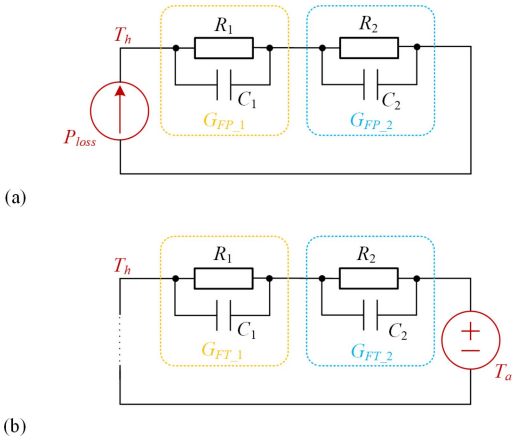


Fig. 3. Equivalent second-order thermal modeling with Foster model with power loss source and ambient temperature source. (a) 2nd-order Foster thermal modeling with power loss source. (b) 2nd-order Foster thermal modeling with ambient temperature source.

The first transfer function G_{CP_1} in Fig. 2(a) can be characterized as:

$$G_{CP_1} = R_2 // \left[1/\left(C_2 s\right)\right] = \frac{R_2}{R_2 C_2 s + 1} \tag{4}$$

where R_2 and C_2 represent the thermal resistance and thermal capacitance of the second layer, respectively.

The second transfer function G_{CP_2} in Fig. 2(a) can be characterized as:

$$G_{CP_2} = \left(G_{CP_1} + R_1\right) // \left[1/\left(C_1 s\right)\right] \tag{5}$$

where R_1 and C_1 represent the thermal resistance and thermal capacitance of the first layer, respectively.

Therefore, the transfer function of Cauer thermal modeling with the power loss source $G^{\text{Cauer}}_{P_{\text{loss}}}(s)$ can be expressed as

$$G^{\text{Cauer}}_{P_{\text{loss}}}(s) = \frac{R_1 R_2 C_2 s + R_1 + R_2}{R_1 R_2 C_1 C_2 s^2 + \left(R_1 C_1 + R_2 C_1 + R_2 C_2\right) s + 1}. \tag{6}$$

Fig. 2(b) illustrates the second-order Cauer thermal modeling with the ambient temperature as the input source. The transfer function from the ambient temperature T_a to the hotspot temperature T_h in the Cauer network is denoted as $G^{\text{Caurer}}_{T_a}(s)$.

The first transfer function G_{CT_1} in Fig. 2(b) can be characterized as

$$G_{CT_1} = \frac{1/\left(C_1 s\right)}{1/\left(C_1 s\right) + R_1}. \tag{7}$$

The second transfer function G_{CT_2} in Fig. 2(b) can be characterized as

$$G_{CT_2} = \frac{\left[1/\left(C_1 s\right) + R_1\right] // \left[1/\left(C_2 s\right)\right]}{\left[1/\left(C_1 s\right) + R_1\right] // \left[1/\left(C_2 s\right)\right] + R_2} \times G_{CT_1}. \tag{8}$$

Therefore, the transfer function of Cauer thermal modeling with the ambient temperature source $G^{\text{Caurer}}_{T_a}(s)$ can be expressed as

$$G^{\text{Caurer}}_{T_a}(s) = \frac{1}{R_1 R_2 C_1 C_2 s^2 + \left(R_1 C_1 + R_2 C_1 + R_2 C_2\right) s + 1}. \tag{9}$$

2) Thermal Modeling Based on Foster Model: Fig. 3(a) illustrates the second-order Cauer thermal modeling with the power loss of the capacitor as the input source.

Similarly, the transfer function of Cauer thermal modeling with the power loss source $G^{\text{Foser}}_{P_{\text{loss}}}(s)$ can be expressed as the sum of the individual transfer functions of each parallel *RC* branch (G_{FP_1} and G_{FP_2})

$$\begin{aligned} G^{\text{Foster}}_{P_{\text{loss}}}(s) &= G_{FP_1} + G_{FP_2} = \left(R_1 // C_1\right) + \left(R_2 // C_2\right) \\ &= \frac{\left(R_1 R_2 C_1 + R_1 R_2 C_2\right) s + R_1 + R_2}{R_1 R_2 C_1 C_2 s^2 + \left(R_1 C_1 + R_2 C_2\right) s + 1}. \end{aligned} \tag{10}$$

In Fig. 3(b), when considering the effect of the ambient temperature source on the hotspot temperature in the Foster thermal network, the transfer function is equal to 1. According to the transfer from the upper part of the thermal network, the transfer function at this point is divided by the sum of the thermal resistances R_1 and R_2 on the basis of G_{FP_1} and transfer function G_{FP_2}. Therefore, the transfer function of Foster thermal modeling with the ambient temperature source $G^{\text{Foser}}_{T_a}(s)$ can be expressed as

$$\begin{aligned} G^{\text{Foser}}_{T_a}(s) &= \frac{G_{FT_1} + G_{FT_2}}{R_1 + R_2} = \frac{\left(R_1 + C_1\right) // \left(R_2 + C_2\right)}{R_1 + R_2} \\ &= \frac{\left(R_1 R_2 C_1 + R_1 R_2 C_2\right) s + R_1 + R_2}{\left[\begin{array}{l} R_1 R_2 C_1 C_2 \left(R_1 + R_2\right) s^2 \\ + \left(R_1 C_1 + R_2 C_2\right)\left(R_1 + R_2\right) s + R_1 + R_2 \end{array}\right]}. \end{aligned} \tag{11}$$

Under constant operating conditions with fixed r_s and R_{th} values, the continuous-frequency (S-domain) thermal model can

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directly compute hotspot temperature from power loss and ambient temperature inputs using its transfer function. However, during long-term dynamic operation, both power loss and ambient temperature become dynamic inputs. Since internal resistance r_s and thermal resistance R_{th} vary with the changing hotspot temperature, the model faces a dual dynamic problem: not only is the input dynamic, but the S-domain transfer function itself also changes simultaneously with temperature. This interdependence makes it infeasible to derive a direct mathematical solution in the continuous-frequency domain, even through iterative discrete calculations. Thus, when modeling long-term dynamic conditions with temperature-dependent r_s and R_{th} variations, obtaining an analytical solution in the S-domain remains highly challenging.

III. Discretized Long Timescale Thermal Modeling

A. Discretized Frequency Domain Thermal Modeling

Considering that the realistic dynamic operating curves are usually discrete data with a set sampling frequency, and the temperature-dependent internal resistance of the capacitor changes dynamically, it is possible to further discretize the continuous frequency-domain transfer model.

The transfer model of a continuous-time system in the S-domain is converted to a discrete-time system in the Z-domain by the Tustin transform ($s \leftarrow [2(1-z^{-1})]/[T(1+z^{-1})]$) [31].

The Z-domain transfer function obtained through the Tustin transform, which characterizes the frequency-domain discrete thermal modeling, can be expressed as

$$g_{P_{\text{loss}}/T_a}^{\text{Cauer / Foster}}(Z) = \frac{A_1 Z^2 + A_2 Z^1 + A_3}{B_1 Z^2 + B_2 Z^1 + B_3} \tag{12}$$

where A_1, A_2, A_3, B_1, B_2, and B_3 represent the coefficients of the discrete thermal network model. The coefficients A_1, A_2, A_3, B_1, B_2, and B_3 in discrete-time model considering the power loss source $g_{P_{\text{loss}}}^{\text{Cauer}}(Z)$ and ambient temperature source $g_{T_a}^{\text{Cauer}}(Z)$ with Cauer thermal parameters are given in (A.1) and (A.2) of the appendix, respectively. Similarly, the coefficients considering the power loss source $g_{P_{\text{loss}}}^{\text{Foster}}(Z)$ and ambient temperature source $g_{T_a}^{\text{Foster}}(Z)$ with Foster thermal parameters are given in (A.3) and (A.4) of the Appendix, respectively.

In order to subsequently perform an iterative analysis of the ESR r_s and hotspot temperature, the discrete Z-domain model is further converted to an iterative model on the effect of power loss on temperature rise $T_{r_p}(t)$ and the effect of ambient temperature on temperature rise $T_{r_a}(t)$

$$\begin{aligned} T_{r_p}(t) = &-\frac{B_2}{B_1}T_{r_p}(t-1) - \frac{B_3}{B_1}T_{r_p}(t-2) \\ &+ \frac{A_1}{B_1}P_{\text{loss}}(t) + \frac{A_2}{B_1}P_{\text{loss}}(t-1) + \frac{A_3}{B_1}P_{\text{loss}}(t-2) \end{aligned} \tag{13}$$

where the gain coefficients are also given in (A.1) and (A.3) of the Appendix (corresponding to Caure and Foster thermal parameters)

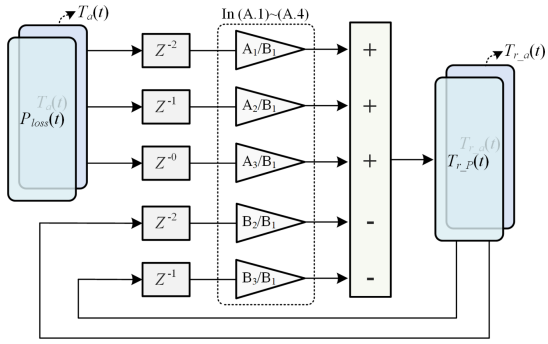


Fig. 4. Calculation flow of temperature rise considering dynamic operation profile based on frequency-domain discrete modeling. (The value of the coefficients A_1, A_2, A_3, B_1, B_2, and B_3 of gain modules are given in (A.1)–(A.4) of the Appendix.)

$$\begin{aligned} T_{r_a}(t) = &-\frac{B_2}{B_1}T_{r_a}(t-1) - \frac{B_3}{B_1}T_{r_a}(t-2) \\ &+ \frac{A_1}{B_1}T_a(t) + \frac{A_2}{B_1}T_a(t-1) + \frac{A_3}{B_1}T_a(t-2) \end{aligned} \tag{14}$$

where the gain coefficients are also given in (A.2) and (A.4) of the Appendix (corresponding to Caure and Foster thermal parameters).

At this point, the temperature rise of the hotspot temperature can be expressed as the sum of the temperature rise of the iterative discrete model with the power loss source and the ambient temperature source. The hotspot temperature $T_h(t)$ in real time can be expressed as

$$T_h(t) = T_{r_p}(t) + T_{r_a}(t) + T_h(0) \tag{15}$$

where $T_h(0)$ indicates the initial temperature of the capacitor, which is equivalent to the initial ambient temperature.

The flow of temperature rise calculation based on frequency-domain discrete modeling with dynamic operating profile is shown in Fig. 4. By solving the delayed iteration module consisting of Z^{-2}, Z^{-1}, and Z^{-0}, and the gain module consisting of A_1, A_2, A_3, B_1, B_2, and B_3, the temperature rise influenced by the power loss source $T_{r-p}(t)$ and the temperature rise influenced by the ambient temperature source $T_{r-a}(t)$ can be obtained, respectively.

B. Temperature-Dependent ESR r_s and Thermal Resistance R_{th}

As shown in Fig. 5, the physical structure of the film capacitors primarily consists of metallized layers and dielectric film layers wound together. The dielectric film layer has a thickness in the millimeter range, while the metallized layer is only nanometers thick. The equivalent internal resistance r_s of the film capacitor is primarily caused by the metallization layer. Due to the influence of thickness, the thermal resistance R_{th} of the film capacitor is mainly determined by the dielectric film layer [32], [33], [34]. The primary medium for thermal resistance from the capacitor external surface to the ambient environment is air. Specifically,

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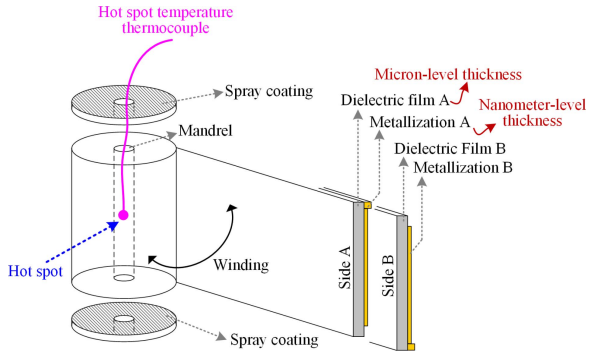


Fig. 5. Physical structure of metallized film capacitors.

for the Cauer model, the thermal resistance of each corresponding layer can be calculated based on the geometric dimensions and thermal conductivity parameters of capacitors with different specifications [35].

While the more commonly used Foster thermal network lacks specific physical meaning, thermal resistance can be indirectly calculated using the temperature rise model. The variation of thermal resistance is analyzed through the temperature rise curve under constant current conditions. Fig. 6(a) illustrates the temperature rise curves for the capacitor with constant 36 A ripple current (100 Hz) over the ambient temperature range of 10 °C–60 °C. This capacitor is one of the experimental samples discussed below. The temperature rise value is calculated as the difference between the hotspot temperature and the case temperature of the capacitor. Based on the temperature rise curves at different ambient temperatures, the relationship between hotspot temperature and temperature rise can be further derived, as shown in Fig. 6(b).

The fitted equation of temperature rise T_{rise} from hotspot to case in Fig. 6(b) is

$$T_{\text{rise}} = 3.375 \times 10^4 \times T_h^{-2.273} + 7.378. \tag{16}$$

Fig. 7(a) gives the measurement results of ESR r_s for this type of capacitors based on *LCR* meter. By measuring the results of r_s at different temperatures and fitting the curve, the corresponding fitted equation is obtained

$$r_s = -1.46 \times T_h^{0.341} + 15.06. \tag{17}$$

By applying the temperature rise calculation formula in Fig. 6(b), the influence of internal resistance can be eliminated. This allows the effect of temperature on thermal resistance R_{th} and its corresponding relationship to be determined, as shown in Fig. 7(b).

The fitting equation for thermal resistance versus temperature in Foster thermal network can be characterized as

$$R_{\text{th}} = 6.806 \times 10^3 \times T_h^{-2.508} + 1.034. \tag{18}$$

From the results, it can be seen that when the hotspot temperature fluctuates within a certain range (from 20 °C–80 °C in this case), both the r_s value and the R_{th} value undergo significant changes. Consequently, these changes in r_s and R_{th} also influence the numerical value of the hotspot temperature. Therefore, the iterative effect between r_s, R_{th}, and hotspot temperature under long timescale operation profiles is not negligible [36], [37], [38].

Furthermore, considering the combined effects of frequency and temperature, measurements of r_s values are taken at different ripple frequencies and temperatures, as shown in Fig. 8. It can be observed that at different temperatures, the variation of r_s with ripple frequency is consistent, first decreasing and then increasing. Conversely, at different frequencies, r_s decreases as temperature increases. Therefore, a unified model to characterize the relationship between r_s and its variation with ripple frequency and temperature can be established

$$\begin{cases} r_s = \left(-1.46 \times T_h^{0.341} + 15.06\right) \\ \quad \times \left(1.34 \times 10^2 \times f_r^{-1.35} + 0.729\right) & (f_r \le 1\text{kHz}) \\ r_s = \left(-1.46 \times T_h^{0.341} + 15.06\right) \\ \quad \times \left(5.84 \times 10^{-7} \times f_r^{1.24} + 0.713\right) & (f_r > 1\text{kHz}). \end{cases} \tag{19}$$

Equation (19) can be applied to different application scenarios to address the requirements for dynamic ambient temperatures over extended time scales and varying ripple frequencies. For instance, in grid-connected renewable energy converter systems, such as wind power and photovoltaic installations, capacitive ripples primarily manifest as low-frequency multiples of the fundamental frequency [5]. Conversely, in electric vehicle and high-speed train traction inverter systems, capacitive ripples predominantly occur as high-frequency multiples of the switching frequency [9], [14]. If capacitors with different specifications are substituted, then the structure of (16)–(19) will remain unchanged, while the coefficients of the fitting equations will change accordingly. Although the new capacitor requires a complete calibration procedure, this requirement remains very feasible in practical industrial applications because manufacturer data can be referenced, and only one measurement conforming to industrial certification standards is needed per capacitor type.

As shown in Fig. 9, the iterative calculation process of r_s and R_{th} and hotspot temperature is exhibited. At time t, calculate the power loss $P_{\text{loss}}(t)$ based on the value of $r_s(t+1)$ and the value of $I_{\text{rms}}(t)$. Simultaneously, use the thermal resistance values R_1, R_2, ... at time t to compute the values of A_1, A_2, A_3, B_1, B_2, and B_3 in (A.1)–(A.4). Then, substitute $P_{\text{loss}}(t)$, the values of A_1, A_2, A_3, B_1, B_2, B_3, as well as the ambient temperature $T_a(t)$ into (13)–(15) to calculate the hotspot temperature $T_h(t)$ at this moment. Based on the hotspot temperature $T_h(t)$, compute the r_s and R_{th} values (including R_1, R_2, etc.) at the next time step $t+1$ using the fitted relationships between internal resistance, thermal resistance, and temperature given in (17)–(19). Iterate the steps above to obtain temperatures, such as $T_h(t+1)$, $T_h(t+2)$, and so on, ultimately yielding the capacitor hotspot temperature result under long-term dynamic operating conditions. Based on this process, thus the hotspot temperature considering the effect of r_s and R_{th} iteration is calculated in real time.

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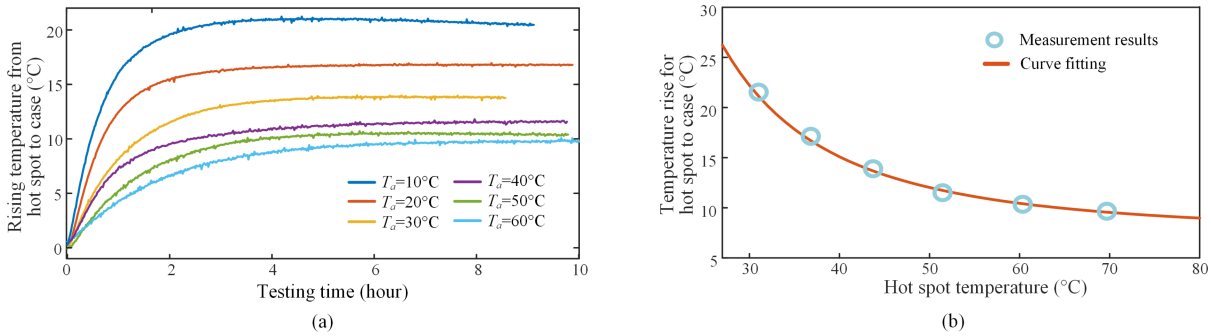


Fig. 6. Measurement and fitting curves of temperature rise at different temperatures. [Maintain a ripple current of 36 A (100 Hz).] (a) Measurement curve of temperature rise at different temperatures. (b) Fitting curve of corresponding temperature rise at different temperatures.

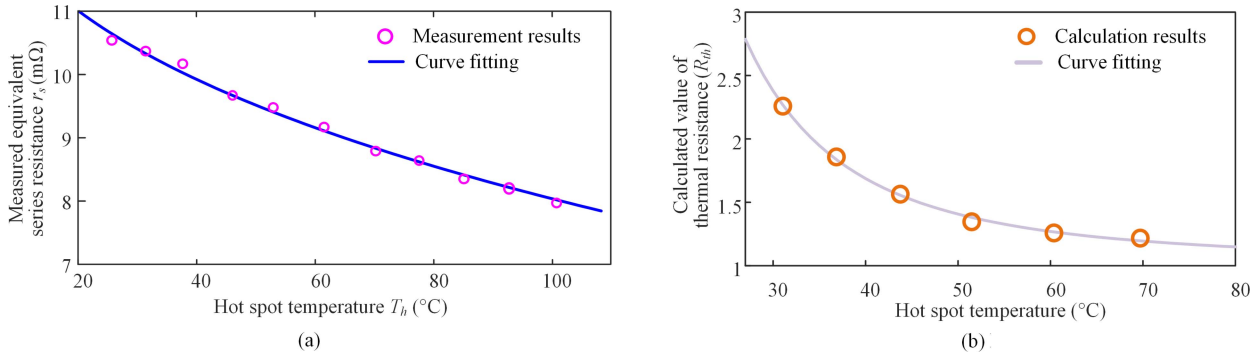


Fig. 7. Fitting curve of corresponding internal resistanc r_s and thermal resistance R_{th} at different temperatures for Foster themal network (r_s measured at 100 Hz by Keysight *LCR* meter E4980 A). (a) Fitting curve of corresponding internal resistance r_s, at different temperatures. (b) Fitting curve of corresponding thermal resistance R_{th} at different temperatures.

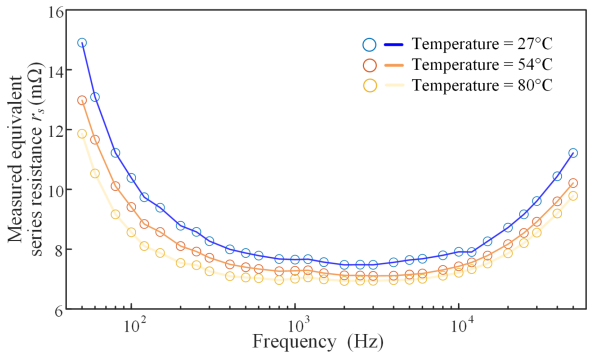


Fig. 8. Fitting curve of corresponding internal resistanc r_s at different frequencies (r_s measured at 100 Hz by Keysight *LCR* meter E4980 A).

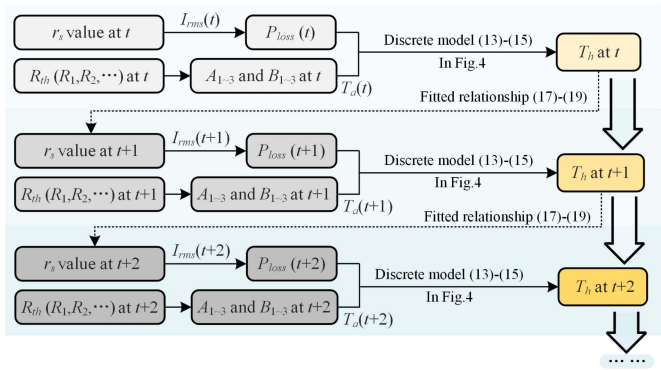


Fig. 9. Iterative computation of r_s, R_{th}, and hotspot temperature.

C. Proposed Long Timescale Thermal Modeling Based on Realistic Mission Profiles

Based on the above iterative analysis of the discretized thermal model, the flowchart of the proposed real-time computation of hotspot temperature on long timescales can be obtained, as shown in Fig. 10. First, the realistic mission profiles of the capacitor are loaded by the operating system, including the ambient temperature $T_a(t)$ and the capacitor current $I_{\text{rms}(t)}$. Meanwhile, the r_s values with the hotspot temperature variations and the thermal network parameters are obtained based on the datasheets or related measurements and calculations. The method in this article is applicable to both the Caure model and the Foster model. Users can choose between the two models based on the information provided by different manufacturers and different types of capacitors. Furthermore, based on the real-time iterative calculations of Figs. 4 and 9 regarding the power loss and ambient temperature, the temperature rise variations of both are thus obtained to get the final hotspot temperature. The hotspot temperature is fed back in real-time to the value of the capacitor internal resistance r_s and thermal reistance r_1, r_2, etc.

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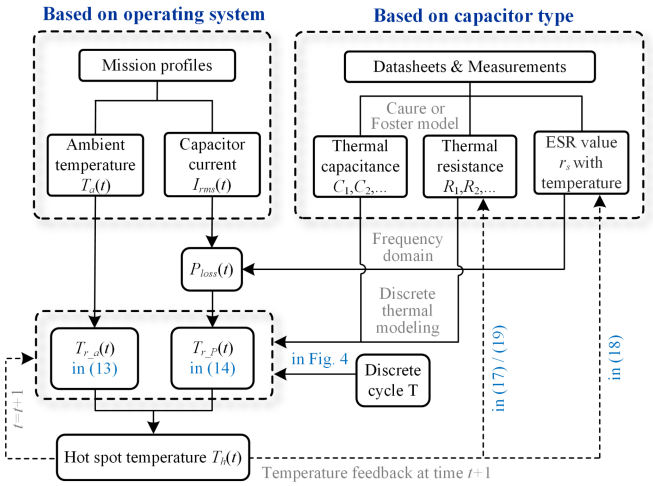


Fig. 10. Procedure for proposed thermal modeling based on long timescale mission profiles.

Therefore, unlike the time-domain thermal model, the frequency-domain thermal model can effectively analyze the joint influence of thermal network parameters on the transfer from power loss to hotspot temperature and from ambient temperature to hotspot temperature. Through discretized iterative computation, the coupled effect of thermal resistance R_{th}, internal resistance r_s, and hotspot temperature can be further derived. Therefore, under long timescale dynamic operating conditions, the frequency-domain thermal model provides higher accuracy in solving for hotspot temperature compared to the time-domain thermal model. Compared to the continuous-frequency model that struggles to simultaneously decouple the temperature-dependent variations of r_s and R_{th}, the establishment of a mathematical model based on the discrete-frequency domain enables a clear calculation of the capacitor hotspot temperature under long-term dynamic operating conditions.

IV. Experimental Verification

A. High-Power Stress Emulation Platform for Capacitors With Dynamic Mission Profiles

The schematic diagram for capacitor thermal stress testing system is shown on Fig. 11. This stress testing architecture preserves the advantages of a recently reported method with a minimum required power supply, and can provide high voltage and current to capacitors under testing (CUTs) [39]. The electrical stress generator is used to apply ripple voltage and ripple current to the CUTs. It generates sinusoidal ripple currents and ripple voltages for the CUT by the voltage supply, the power stack for three-phase inverter, and the AC-side inductor.

The controller uses a three-phase space vector pulsewidth modulation scheme to modulate the three-phase current with a sinusoidal waveform. In the 3/2 coordinate transformation, the reference value of the two-phase rotation coordinates, D-axis component can be set to the mission profiles for amplitude of the ripple current, Q-axis component can be set to 0, and the rotation

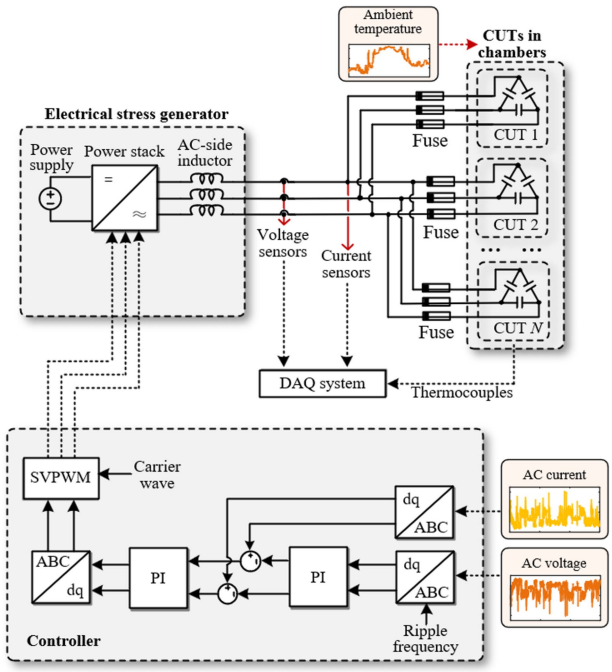


Fig. 11. Schematic diagram for capacitor thermal stress testing and DAQ system.

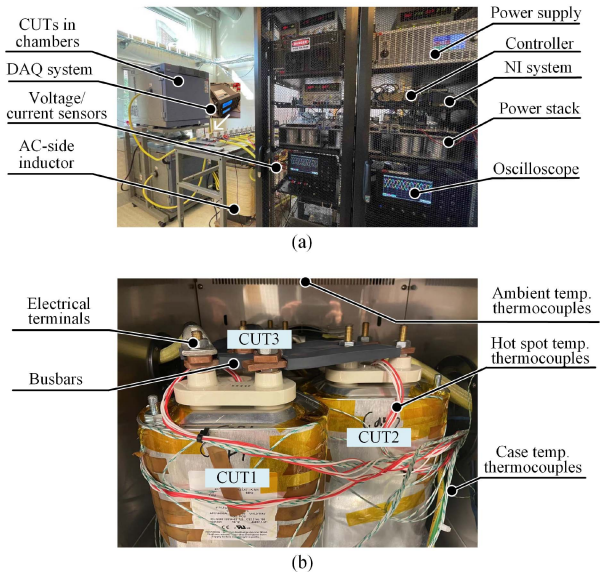


Fig. 12. Experimental platform for capacitor stress emulation and DAQ. (a) Photo of stress emulation and data acquisition equipment. (b) CUTs in the chamber.

angle can be set to the ripple frequency of the CUTs [40]. The programmable thermostat chambers can emulate the ambient temperature of CUTs by importing the realistic temperature mission profile. Meanwhile, the electrothermal stresses are collected in real-time by data acquisition (DAQ) systems [41].

The experimental platform for capacitor stress emulation and DAQ is shown in Fig. 12. The capacitors are equipped

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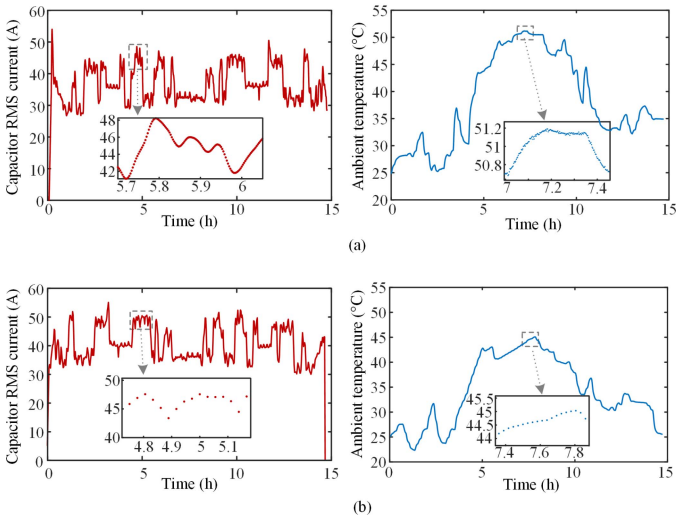


Fig. 13. Dynamic operating profile profiles for two cases (Mission profiles in two cases set up different sampling intervals). (a) Mission Profile for Case 1 (Sampling interval time is 10 s). (b) Mission Profile for Case 2 (Sampling interval time is 100 s).

with thermocouples measuring hotspot temperature and case temperature, while an ambient temperature thermocouple is positioned at the center of the test chamber. The hotspots are obtained in real-time by burying the thermocouple at the center core temperature location. As shown in Fig. 5, the hotspots temperature point in film capacitors is located at the center point of the cylindrical physical structure [32], [33], [34].

Taking one type of capacitor operating condition for AC filtering in the traction system as an example, Fig. 13 gives the mission profiles for two cases, including the capacitor current and the ambient temperature in the converter system [42]. The sampling frequency should be at least 10–15 times faster than the fastest thermal time constant [43]. Different sampling intervals time of 10–100 s are set for the mission profiles in the two cases, respectively.

B. Experimental Results

Initial parameters of r_s and R_{th} can be obtained by the hotspot temperature ramp-up-down curves measured under the constant-current test, which is shown in the results of Figs. 6 and 7. The relationship between r_s and temperature can be obtained by offline measurement of *LCR* meter. Meanwhile, the second-order thermal network parameters, including C_1, C_2, R_1, and R_2, are obtained by fitting the Foster model. The configuration of the experimental platform and the relevant parameters of the capacitor are shown in Table II.

According to the flowchart in Fig. 10, the calculated hotspot temperature rise due to self-heating by ambient temperature and power loss can be obtained in real-time as given in Fig. 14. From the results of the two cases, it can be seen that the calculated temperature rises $T_{r_a}(t)$ and $T_{r_p}(t)$ are smoother compared to the changes in the mission profiles, influenced by the transfer function of the thermal model.

TABLE II
TESTING SYSTEM AND CAPACITOR PARAMETERS

Stress emulation setup				
DC power supply voltage U_{PS}	750 V			
AC-side inductor L	2.5 mH			
Controller switch frequency f_S	4000 Hz			
CUTs setup for AC capacitors				
Capacitance of the CUT unit	75 μF			
Testing AC current (rms) I_{rms}	0-60 A			
Ripple frequency f_{test}	100 Hz			
Testing ambient temperture T_a	25 °C - 55 °C			
ESR r_s	10.4 mΩ (30°C)			
Thermal parameter (30°C)	R_1 (°C/W)	R_2 (°C/W)	C_1 (J/°C)	C_2 (J/°C)
	2.24	0.24	2980	28900

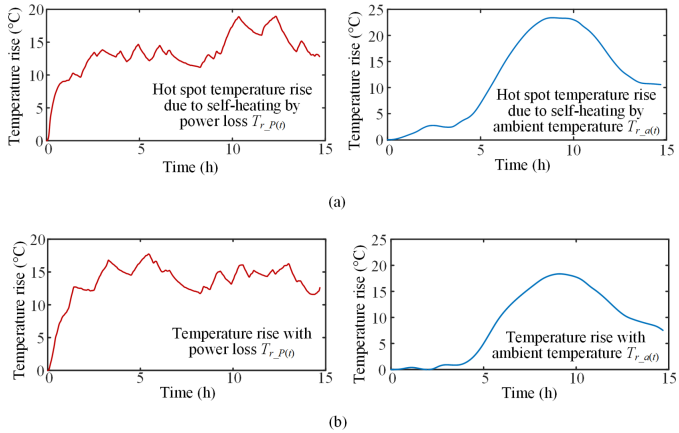


Fig. 14. Hotspot temperature rise due to self-heating by ambient temperature and power loss in two cases. (a) Hot spot temperature rise due to self-heating by ambient temperature and power loss in Case 1. (b) Hot spot temperature rise due to self-heating by ambient temperature and power loss in Case 2.

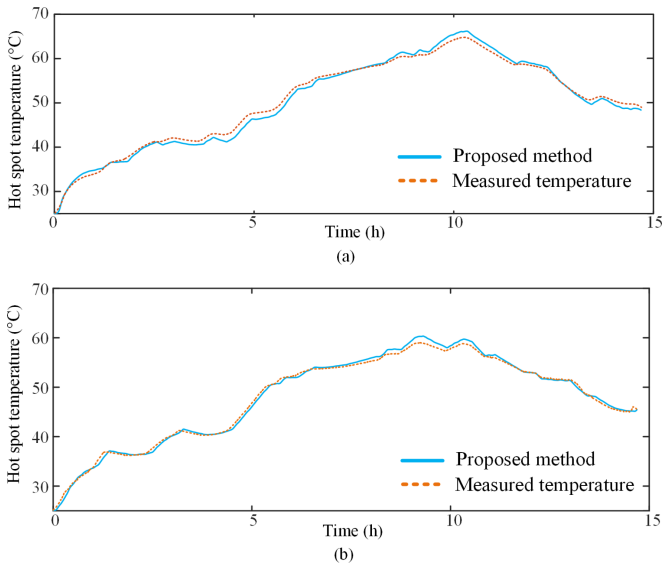


Fig. 15. Hotspot temperature obtained from thermal modeling calculations in two cases. (a) Hot spot temperature obtained from thermal modeling calculations in Case 1. (b) Hot spot temperature obtained from thermal modeling calculations in Case 2.

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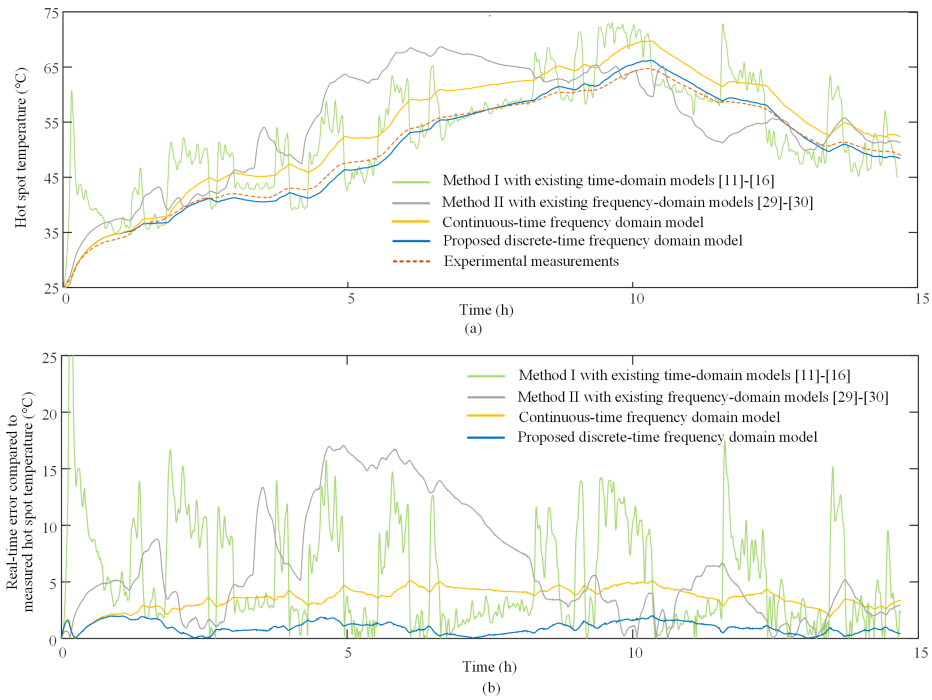


Fig. 16. Hotspot temperatures and errors from different methods on long timescales. (a) Calculated hot spot temperature results using different methods and measured hotspot temperature results. (b) Comparison of calculation errors in hot spot temperature using different methods.

Finally, by superimposing the two temperature rises $T_{r_a}(t)$ and $T_{r_p}(t)$, the calculation results of the hotspot temperature are obtained. Fig. 15 illustrates the comparison between the calculated hotspot temperatures based on the proposed method and measured temperatures with the two mission profile cases. From the results, the trend of the results calculated by the proposed method and the measured hotspot temperature are consistent, and the maximum error is less than 2 °C. This indicates that the proposed method is reasonable for dynamic operational profiles at different sampling interval times.

C. Comparison With Existing Methods

1) Comparison of Hotspot Temperature Calculation Results: Taking Case 1 as an example, the hotspot temperature calculation results of the proposed method are compared with existing methods, which are given in Fig. 16. Method I characterizes the type of existing time-domain thermal calculation models [11], [12], [13], [14], [15], [16]. Method II characterizes the type of existing frequency-domain thermal calculation models [29], [30]. In addition, the results of calculations that the continuous-time frequency domain model and the proposed discrete-time frequency domain model are compared.

From the results, it is seen that when considering long timescales and multioperating mission profile inputs, the error of the conventional method is relatively large, and does not match the trend of the realistic hotspot temperature. In the proposed method, the calculation results that consider the effect of the temperature variation of the capacitor internal resistance r_s are significantly closer to the measured hotspot temperature than those that do not consider its effect. The maximum hotspot temperature of the existing time-domain model is obtained at about 6 h, which is a significant difference. The offset of the maximum hotspot temperature point leads to a large error in the existing time-domain model at long timescales, especially in the middle time period. Due to the existing time-domain model, due to the limitation of the thermal network for the dynamic external temperature processing, the change of its ambient temperature will be directly superimposed on the hotspot temperature, which leads to the main error. Meanwhile, the error of the existing frequency domain model is related to the dynamics of the capacitor ripple current. When the ripple current is in steady state, the error of the existing frequency domain model is small at this time. However, when the ripple current changes, the error of the existing frequency domain model gradually increases. Therefore, existing time-domain and frequency-domain methods still exhibit limitations when operating under capacitor dynamic current conditions in long timescale operations. In addition, as the capacitor operation over extended time scales, the hotspot temperature calculated by the frequency-domain model under the continuous transfer function becomes significantly higher than the measured temperature. In contrast, the temperature calculated by the frequency-domain model under the discrete transfer function remains consistent with the measured temperature.

The detailed errors for different methods are summarized in Fig. 17. For hotspot temperatures of high-power capacitors with long timescales and multiple operating conditions, the maximum and average errors of conventional methods exceed 15 °C–5 °C, respectively. The error of the proposed method is obviously lower, especially considering the dynamic effect of the capacitor internal resistance r_s, its maximum and average errors are only 1.9 °C–0.92 °C, respectively.

2) Comparison of Lifetime Expectancy Calculation Results: Accurate acquisition of hotspot temperatures is critical to the

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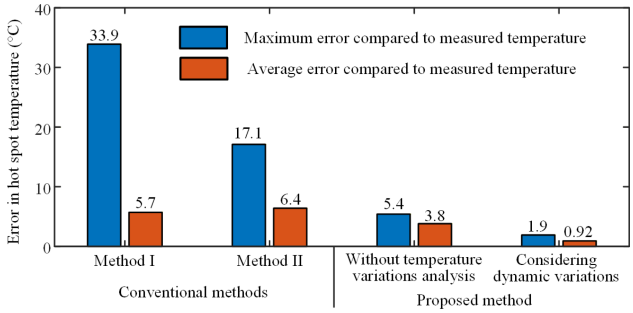


Fig. 17. Errors in the hotspot temperatures calculated by the different methods compared to the measured values (Compared calculation methods are consistent with Fig. 16).

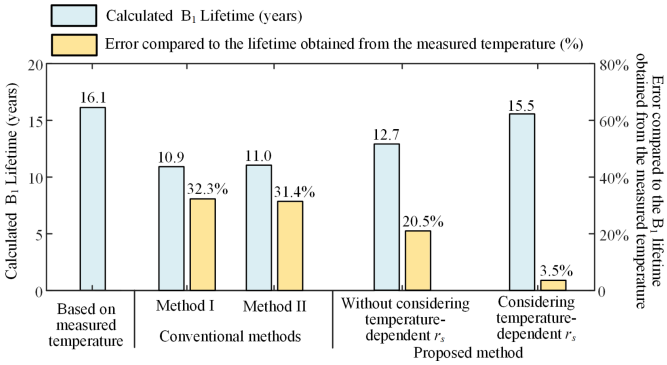


Fig. 18. B_1 lifetime calculation results with different hotspot temperature models. (B_1 lifetime calculation based on one type of representative film lifetime model and unreliability Weibull distribution; Compared calculation methods are consistent with Fig. 16).

calculation of capacitor lifetime expectancy [2], [44]. The lifetime expectancy obtained from different hotspot temperature calculation methods is further discussed.

One representative film capacitor lifetime model can be characterized as [2]

$$L = L_0 \times \left(\frac{U}{U_0}\right)^{-p} \times 2^{\frac{T_0 - T_h}{10}} \tag{20}$$

where L, L_0, U, U_0, and T_0 denote the lifetime expectancy, the rated lifetime, the operation voltage, the rated voltage, and the rated temperature, respectively. p is an empirical coefficient, and a typical value for film capacitors is 4 [2], [45].

The operation voltage U and hotspot temperature T_h are obtained by running the mission profiles for Case 1, and cumulatively calculating the lifetime according to the Miner linear damage theory [46]. The accumulated failure probability is further calculated using the Weibull distribution. One case study shows a typical value of 5.06 for the shape parameter in the Weibull distribution of film capacitor failures in traction systems [14]. Based on the results of the life calculations in (20), the B_1 lifetime under the unreliability distribution of the capacitor can be obtained, i.e., it means that 1% of the samples are expected to fail at that time. The B_1 lifetime calculated based on different hotspot temperature modeling methods is shown in Fig. 18. The lifetime expectancy obtained by conventional hotspot temperature calculations has an error of about 30% compared to that based on measured temperatures. The lifetime expectancy obtained by the hotspot temperature calculation in the proposed method is closer to the result obtained from the measured temperature, with an error of only 3.5%. This demonstrates that the hotspot temperature calculation method proposed in this article benefits in the meantime to obtain precise lifetime expectancy.

V. Conclusion

This article proposes a hotspot temperature model for film capacitors, which applies to high-power converter systems with long timescale dynamic operation profiles. The effects of power loss and ambient temperature are separately characterized as frequency-domain transfer functions with thermal parameters. The transfer function is discretized to obtain the real-time temperature rise influenced by power loss and ambient temperature, allowing discrete operation profiles to be conveniently introduced into the temperature model. Furthermore, the temperature-dependent ESR and thermal resistance can be iteratively calculated in frequency-domain modeling. The rationality and accuracy of the proposed method is verified by a full-power experimental platform for capacitor stress testing. From the hotspot temperature results, the maximum and average errors of conventional methods exceed 15 °C and 5 °C, respectively. The error of the proposed method is obviously lower, its maximum and average errors are only 1.9 °C and 0.92 °C, respectively. The lifetime expectancy obtained by conventional hotspot temperature calculations has an error of about 30% compared to that based on measured temperatures. The lifetime expectancy obtained by the hotspot temperature calculation in the proposed method is closer to the result obtained from the measured temperature, with an error of only 3.5%. This demonstrates that the calculation method proposed in this article benefits in obtaining precise hotspot temperature and lifetime expectancy, thus contributing to the initial design and safe operation of capacitors.

Appendix A
Correlation Coefficients for Discretized Frequency-Domain Thermal Models

A_1, A_2, A_3, B_1, B_2, and B_3 represent the discrete coefficients in (12) and the gain coefficients in (13) and (14).

The discrete coefficients for frequency domain model $g_{P_{\text{loss}}}^{\text{Caure}}(Z)$ in (12) and gain coefficients for iterative model $T_{r_p}(t)$ in (13) with Caure thermal parameters by the power loss source are given as

$$\begin{cases} A_1 = R_1T^2 + R_2T^2 + 2R_1R_2C_2T \\ A_2 = 2R_1T^2 + 2R_2T^2 \\ A_3 = R_1T^2 + R_2T^2 - 2R_1R_2C_2T \\ B_1 = T^2 + 2R_1C_1T + 2R_1C_2T + 2R_2C_2T + 4R_1R_2C_1C_2 \\ B_2 = 2T^2 - 8R_1R_2C_1C_2 \\ B_3 = T^2 - 2R_1C_1T - 2R_2C_1T - 2R_2C_2T + 4R_1R_2C_1C_2 \end{cases} \tag{A.1}$$

where T denotes discrete time intervals.

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The discrete coefficients for frequency domain model $g_{T_a}^{\text{Caure}}(Z)$ in (12) and gain coefficients for iterative model $T_{r_a}(t)$ in (14) with Caure thermal parameters by the ambient temperature source are given as

$$\begin{cases} A_1 = T^2 \quad A_2 = 2T^2 \quad A_3 = T^2 \\ B_1 = T^2 + 2R_1C_1T + 2R_2C_1T + 2R_2C_2T + 4R_1R_2C_1C_2 \\ B_2 = 2T^2 - 8R_1R_2C_1C_2 \\ B_3 = T^2 - 2R_1C_1T - 2R_2C_1T - 2R_2C_2T + 4R_1R_2C_1C_2. \end{cases} \tag{A.2}$$

The discrete coefficients for frequency-domain model $g_{P_{\text{loss}}}^{\text{Foster}}(Z)$ in (12) and gain coefficients for iterative model $T_{r_p}(t)$ in (13) with Foster thermal parameters by the power loss source are given as

$$\begin{cases} A_1 = R_1T^2 + R_2T^2 + 2R_1R_2C_1T + 2R_1R_2C_2T \\ A_2 = 2R_1T^2 + 2R_2T^2 \\ A_3 = R_1T^2 + R_2T^2 - 2R_1R_2C_1T - 2R_1R_2C_2T \\ B_1 = T^2 + 2R_1C_1T + 2R_2C_2T + 4R_1R_2C_1C_2 \\ B_2 = 2T^2 - 8R_1R_2C_1C_2 \\ B_3 = T^2 - 2R_1C_1T - 2R_2C_2T + 4R_1R_2C_1C_2. \end{cases} \tag{A.3}$$

The discrete coefficients for frequency domain model $g_{T_a}^{\text{Foster}}(Z)$ in (12) and gain coefficients for iterative model $T_{r_a}(t)$ in (14) with Foster thermal parameters by the ambient temperature source are given as

$$\begin{cases} A_1 = R_1T^2 + R_2T^2 + 2R_1R_2C_1T + 2R_1R_2C_2T \\ A_2 = 2R_1T^2 + 2R_2T^2 \\ A_3 = R_1T^2 + R_2T^2 - 2R_1R_2C_1T - 2R_1R_2C_2T \\ B_1 = R_1T^2 + R_2T^2 + 2R_1^2C_1T + 2R_2^2C_2T + 2R_1R_2C_1T \\ \quad +2R_1R_2C_2T + 4R_1R_2^2C_1C_2 + 4R_1^2R_2C_1C_2 \\ B_2 = 2R_1T^2 + 2R_2T^2 - 8R_1^2R_2C_1C_2 - 8R_1R_2^2C_1C_2 \\ B_3 = R_1T^2 + R_2T^2 - 2R_1^2C_1T - 2R_2^2C_2T - 2R_1R_2C_1T \\ \quad -2R_1R_2C_2T + 4R_1R_2^2C_1C_2 + 4R_1^2R_2C_1C_2. \end{cases} \tag{A.4}$$

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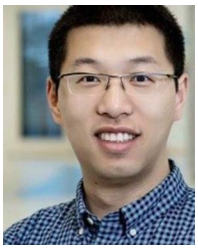
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